

Momentum

$$p = mv$$

Momentum = Mass * Velocity

Express Newton 2nd law in terms of momentum

$$F = ma$$

$$= m \times \frac{\Delta v}{t}$$

$$= \frac{mv_i - mv_f}{t}$$

$$= \frac{\Delta p}{t}$$

Total Momentum

$$\Sigma P = P_i + P_j$$

Conservation of Momentum

Total Momentum for a system = 0

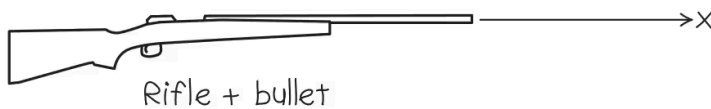
Total Momentum → Depends on the external force

Condition

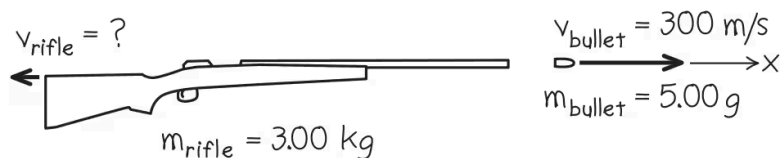
- No external force

Example for Recoil on a Rifle

Before



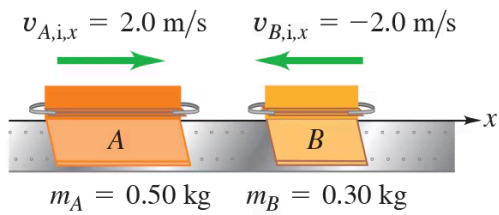
After



- Rifle is in contact with the bullet
- To solve the velocity of this bullet

$$\begin{aligned}\Sigma p &= 0 = (5 \times 10^{-3})(300) + (3)(v_{rifle}) \\ &= -0.5 \text{ ms}^{-1}\end{aligned}$$

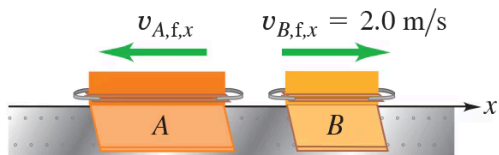
★ Must Review for Midterm



(a) Before collision



(b) Collision



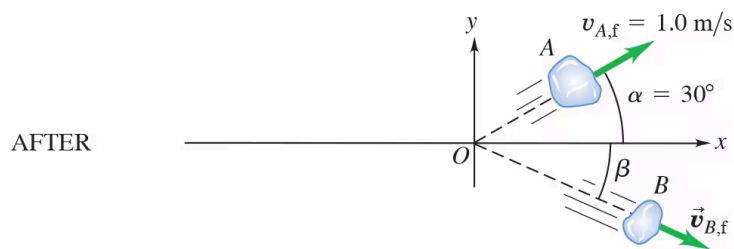
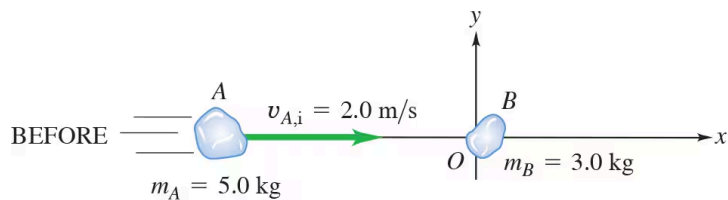
(c) After collision

Initial Momentum = Final Momentum

$$p_{\text{initial}} = (0.5)(2) + (0.3)(-2)$$

$$p_{\text{final}} = (0.5)(v) + (0.3)(+2)$$

$$v = -0.4 \text{ ms}^{-1}$$



X Component:

$$\text{initial momentum} = 2 * 5 + 0 = 10$$

$$\text{final momentum} = 5 * 1 * \cos 30 + 3 * v$$

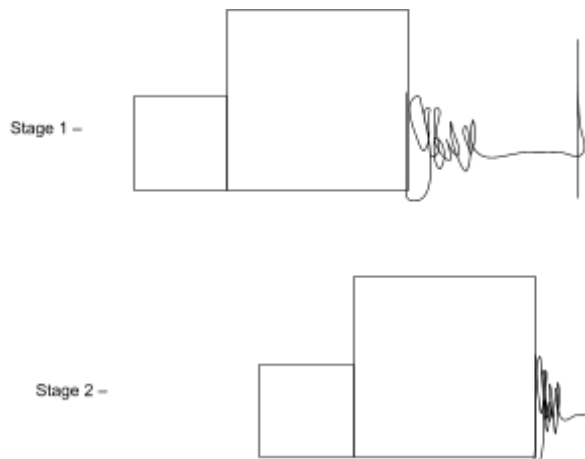
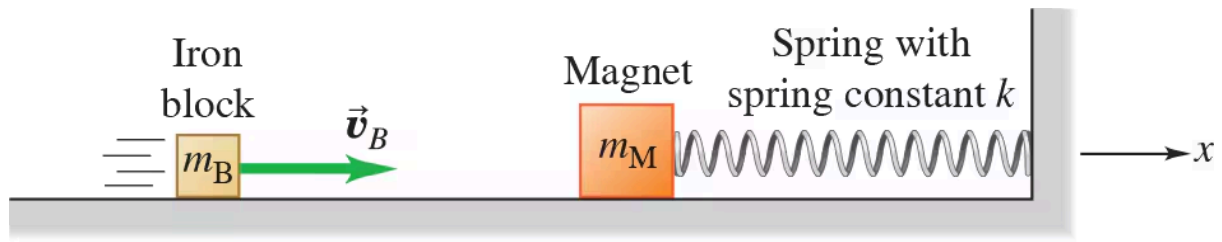
Y Component

$$\text{initial} = 5 * 0 + 3 * 0 = 0$$

$$\text{final} = 5 * 1 * \sin 30 + 3 * v = 0 \leftarrow \text{To find the y component of particle B}$$

Elastic vs Inelastic Collision

- Elastic – Kinetic Energy MUST be conserve



Before Compression → KE only

After Compression → Spring Energy

K_E In terms of Momentum

$$KE = \frac{1}{2}mv^2 \quad p = mv \rightarrow v = \frac{p}{m}$$

$$KE = \frac{1}{2}m\left(\frac{p}{m}\right)^2 \rightarrow = \frac{1}{2}\frac{p^2}{m}$$

$$KE = \frac{p^2}{2m}$$